
INSTREAM POST-IMPLEMENTATION TIMESERIES UTILITY USER DOCUMENT

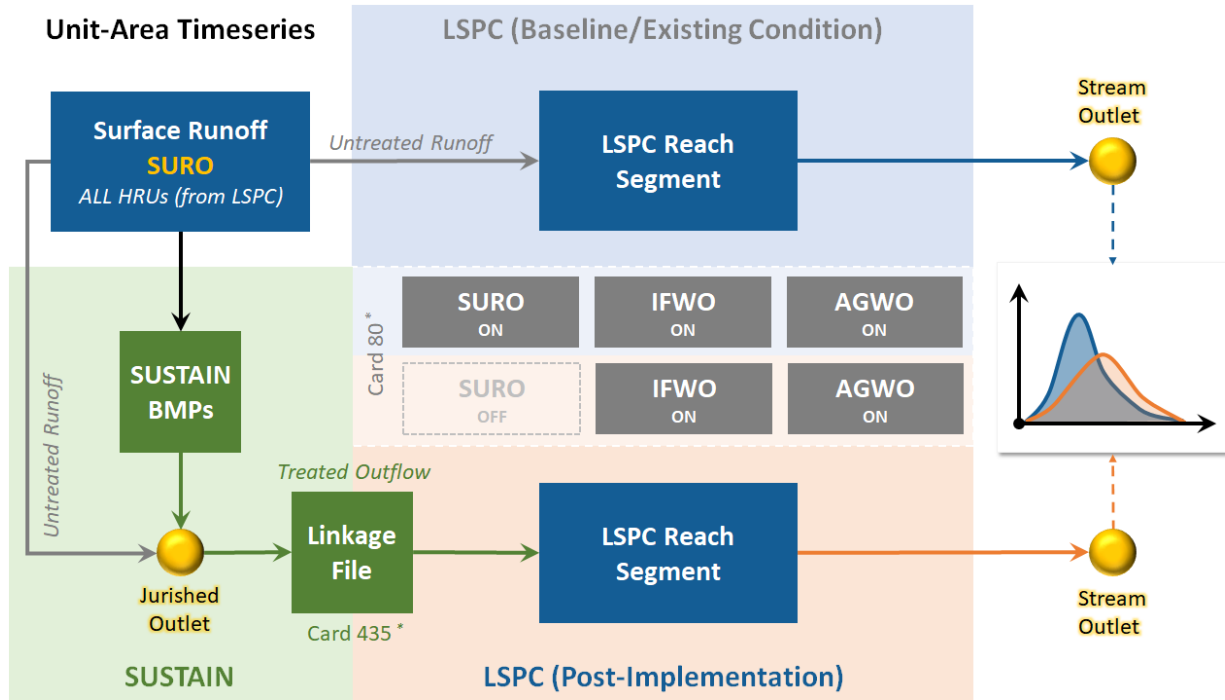
The Instream Post-implementation Timeseries Utility (Utility) provides a pathway to link output from the Optimization Framework to the calibrated WMMS 2.0 LSPC model. Once a user has successfully executed the WMMS 2.0 Optimization Framework, they can select a suite (i.e., “slice”) of BMPs that meets the intended percent reduction of pollutant loads or volume from the curve on the Optimization Viewer. They can also choose to create an implementation plan from the Optimization Viewer. An implementation plan details the selected BMP type, size, reduction, and cost for each modeled watershed. This Utility simulates the hydrologic and water quality impact the selected BMPs provide in LSPC. It generates a batch file that executes a series of SUSTAIN runs, which result in an alternate set of flow and pollutant load timeseries at the most downstream outlet of each watershed. Those timeseries can be incorporated into the calibrated LSPC model, replacing the existing condition edge-of-stream runoff and pollutant loads, and enabling users to evaluate BMP impacts on instream pollutant concentrations. This document describes the inputs, processing, and outputs for generating post-implementation timeseries with this Utility.

1 UTILITY DESCRIPTION

LSPC has an optional feature to create “linkage files” through the configuration of *Card 435* in the input file. A linkage file is a type of configured timeseries file. This option allows for the segmentation of the watershed model at points in the reach network as dictated by a user. One application of this functionality is to divide large models that may require excessive runtime or hardware resources into a series of separate, sequentially linked model runs. For example, in a situation with undeveloped headwaters upstream of a developed area where management scenarios will be evaluated, the model may only need to be run one time for the headwaters if no changes are made during the management scenarios. In this example, the headwater portion of the model could be run and the results written to one or more linkage files for use as a boundary condition to the downstream portion of the model where several different management scenarios are evaluated. The application of this linkage feature provides model runtime efficiencies, especially for large watershed networks.

In context of this Utility, linkage files are leveraged to integrate model results from SUSTAIN back into the calibrated LSPC model to represent post-implementation scenarios. This Utility generates a batch file that a user can use to execute a series of SUSTAIN runs to generate an hourly output timeseries representing the post-BMP implementation runoff and pollutant loading. This output will be in the format of an LSPC linkage file, which can then be directly configured into an LSPC run. LSPC accepts linkage files through model *Card 435 – Optional Linkage Controls*. One linkage file is required for the outlet of each corresponding watershed from SUSTAIN to be fully represented in LSPC. LSPC allows for more than one linkage file to be routed to any given reach segment, making it possible to aggregate higher-resolution watershed outputs up to the parent subwatershed/reach scale.

The linkage file replaces the runoff (i.e., SURO) component of the baseline watershed model. To avoid double counting that contribution, the user must disconnect the modeled SURO timeseries from the baseline model for every subwatershed where a linkage file from SUSTAIN is being represented. This is done by setting all the HRU-level values for ROUTE_SURO in *Card 80* to a value of -1. Because *Card 80* is generalized by DEFID instead of subwatershed, configuration changes are applied to ALL subwatersheds associated with the DEFID for which ROUTE_SURO has been disconnected. Figure 1 is a schematic of the post-implementation timeseries utility workflow and model linkages.



* SUSTAIN-to-LSPC linkage occurs at two key points. Card 435 assigns SUSTAIN jurished outlet timeseries to corresponding LSPC Reach segments. LSPC will still always simulate the full water balance; however, for the Post-Implementation scenario, Card 80 must be changed to disconnect the baseline SURO being replaced by SUSTAIN timeseries.

Figure 1. Schematic of post-implementation timeseries utility workflow and model.

It is generally recommended that the model be configured such that 100 percent of the jurished area is translated to the SUSTAIN environment, with areas that are not subject to management being routed directly to the jurished outlet as “Untreated.” This approach provides several advantages for this Utility. It simplifies the linkage because 100% of SURO can be disconnected in LSPC and replaced by the linkage file, preserving flow and mass balance of both treated and untreated areas. If this approach is not followed, then the user is responsible to ensure that the HRU area table in LSPC (Card 90, specified for every subwatershed and HRU combination) has been adjusted accordingly to exclude treated areas that are being replaced by SUSTAIN linkage files.

2 UTILITY EXECUTION

Execution of the Utility begins by uploading a *.crz file from the Two-tiered Optimization Utility to the Optimization Viewer. The Viewer summarizes the complete range of load or flow reduction from low to high on the x-axis and the corresponding capacity for each BMP type (primary y-axis) and cost (secondary y-axis) in a cumulative distribution plot (Figure 2).

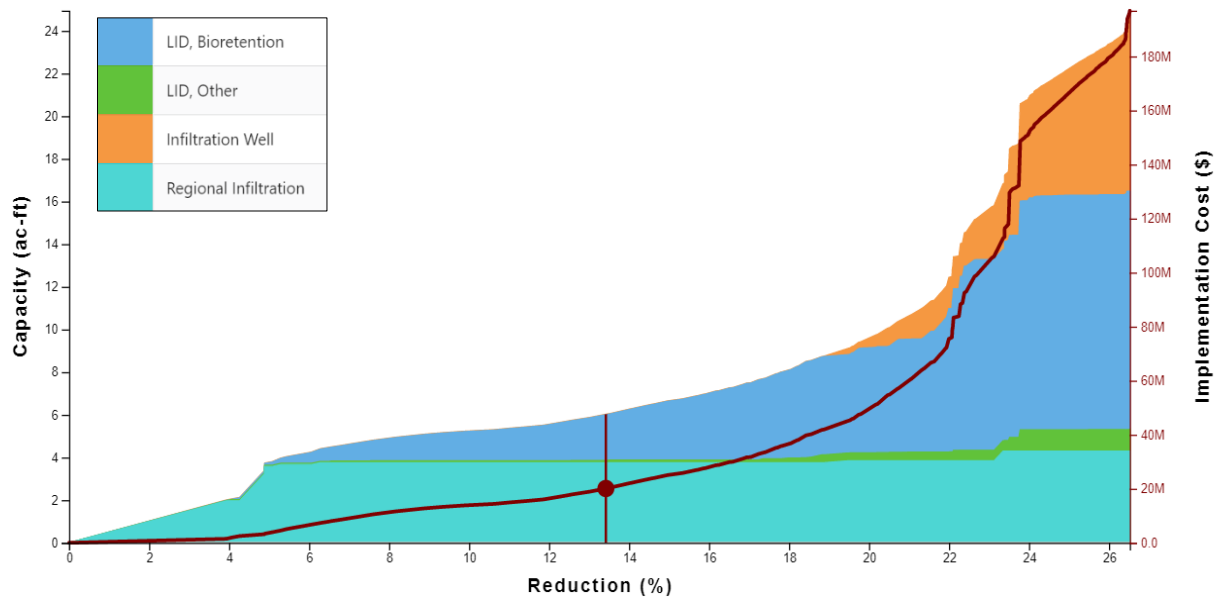


Figure 2. Cumulative load reduction summary plot with corresponding BMP types, volumes, and costs.

Once a percent reduction has been selected from the Optimization Viewer, the corresponding suite of BMPs can be organized into SUSTAIN inputs files for simulation at each jurisdiction by clicking the 'ToLSPC' button (Figure 3). This will execute the Post-implementation Timeseries Utility.

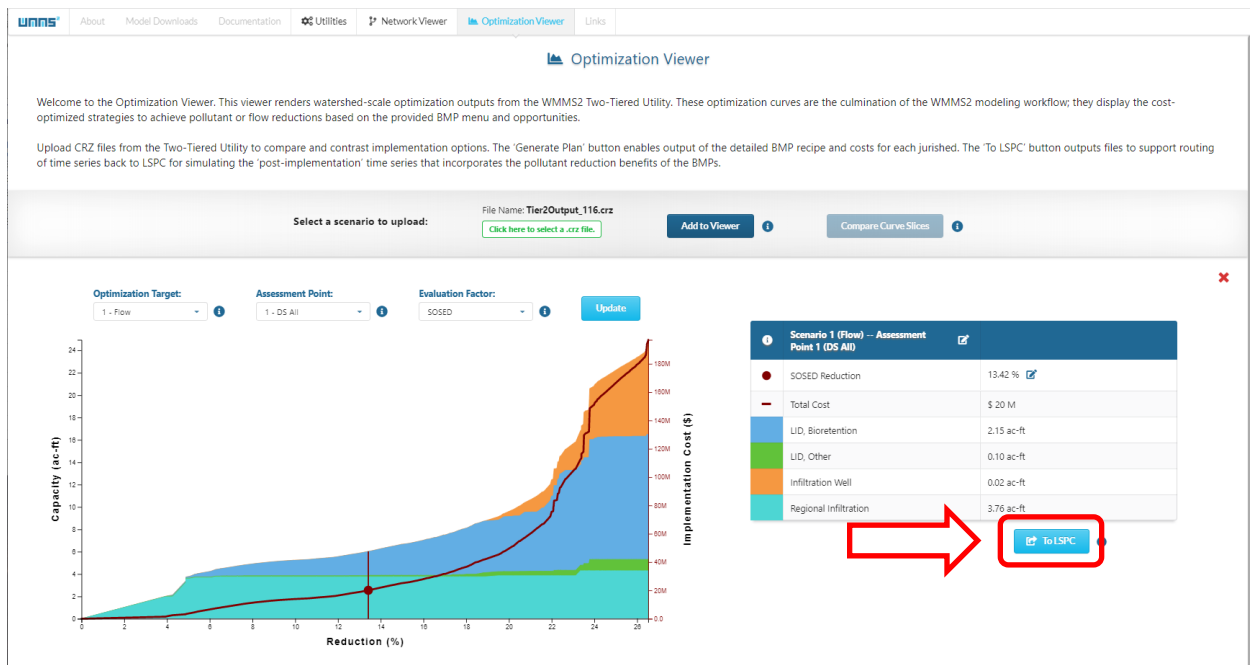


Figure 3. Optimization Viewer with a suite of BMPs selected and the 'ToLSPC' button highlighted.

The Utility generates a batch file (s2l.bat) that runs a series of SUSTAIN input files with the selected 'slice' from the curve with the applicable BMPs selected and sized accordingly. The commands within the batch file will execute each jurisdiction's input file to generate an hourly, post-implementation timeseries in the format of the LSPC linkage file. Prior to execution, the batch file should be located on the user's local drive as shown in Figure 4 (C:\MyFiles\WMMS\SUSTAIN\Current\2production).

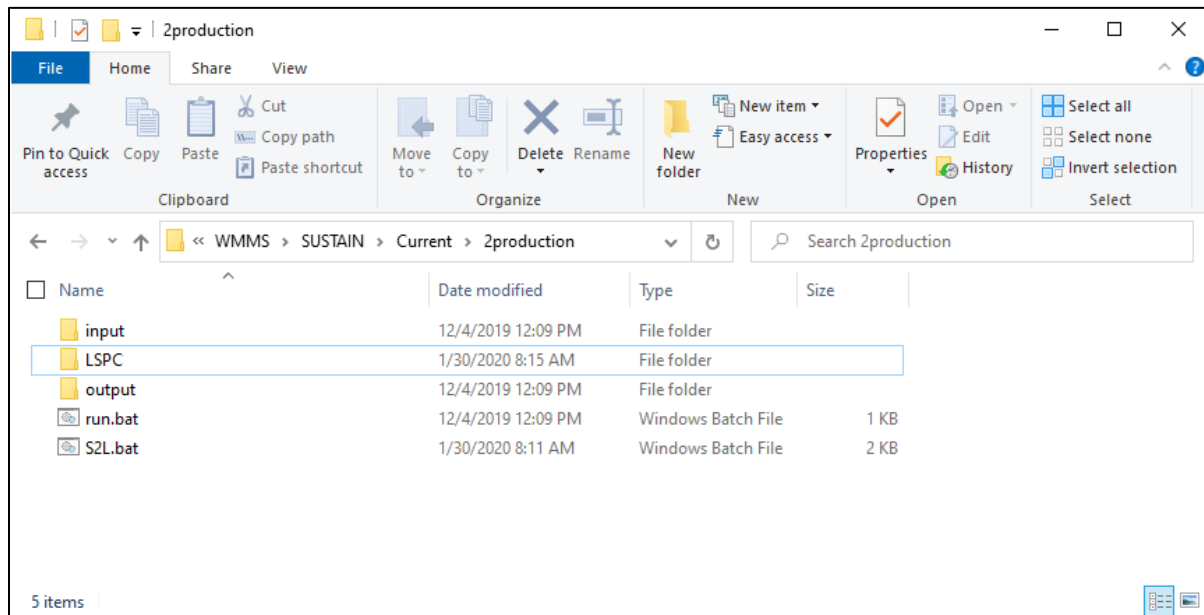


Figure 4. File directory where output from the Utility should be unzipped.

Depending on the number of jurisdictions, timeseries length, and the complexity of the routing network, this process may take several hours. When completed, the SUSTAIN-generated output timeseries files with *.inp extensions will be located on the user's local folder directory as shown in Figure 5 (C:\MyFiles\WMMS\SUSTAIN\Current\2production\LSPC).

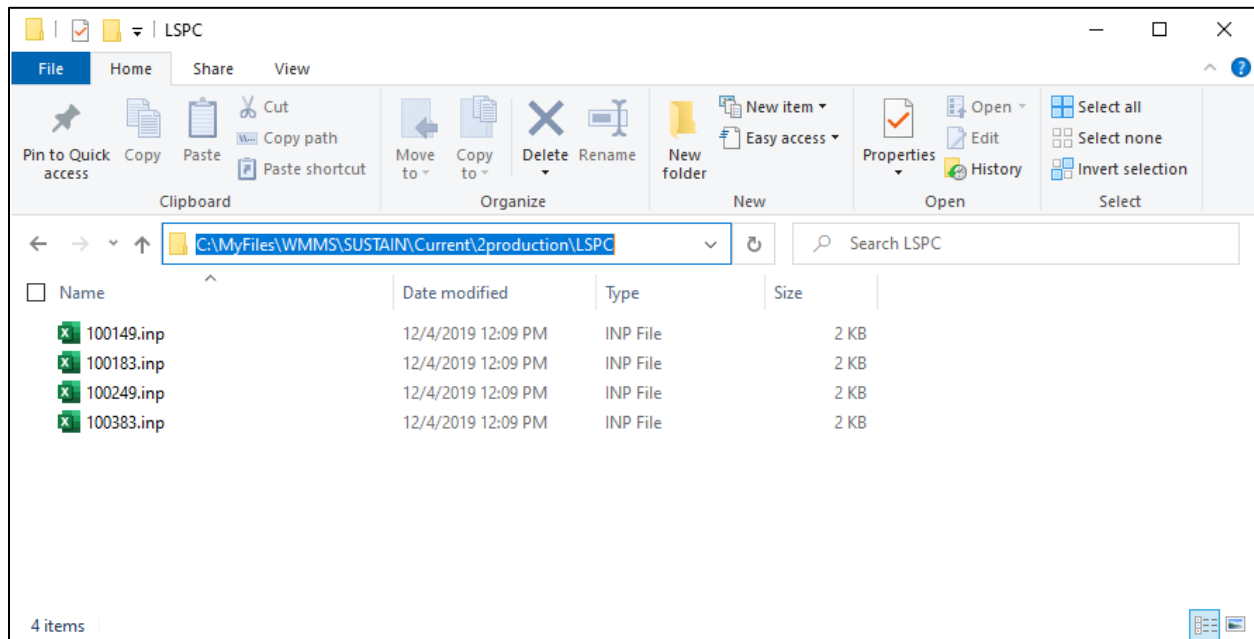


Figure 5. File directory where output from the SUSTAIN timeseries simulation will be exported.

After the SUSTAIN timeseries files have been generated, *Card 435 – Optional Linkage Controls* in the corresponding LSPC model input file must be configured to read in these timeseries. Figure 6 shows how each *.inp from Figure 5 is read into the corresponding subbasin/subwatershed (e.g., jurisdiction 100383 corresponds to subwatershed 1003). All outlet flags are '0' in this type of linkage configuration.

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File Edit Format View Help
c435 linkage controls (optional)
c
c      rchid      reach id (same as subbasin id)
c      outlet     outlet control switch for the corresponding reach
c                0 = will read input timeseries file
c                n = will write output timeseries file for outlet n
c      filepath   the timeseries file path (continuous string without space)
c
c rchid      outlet      filepath
1003         0           C:\MyFiles\WMMS\SUSTAIN\Current\2production\LSPC\100383.inp
1001         0           C:\MyFiles\WMMS\SUSTAIN\Current\2production\LSPC\100149.inp
1001         0           C:\MyFiles\WMMS\SUSTAIN\Current\2production\LSPC\100183.inp
1002         0           C:\MyFiles\WMMS\SUSTAIN\Current\2production\LSPC\100249.inp
```

Figure 6. Example configuration of LSPC Card 435 – Optional Linkage Controls.

Likewise, assuming the linkage files replace 100 percent of the SURO edge-of-stream timeseries from the baseline model run, the ROUTE_SURO value in *Card 80 – Land Use to Stream Routing* for each DEFID and HRU combination must be set to a value of “-1” to disconnect the existing-condition edge-of-stream SURO timeseries from the model. Those timeseries are replaced by the post-implementation timeseries from SUSTAIN. The ROUTE_IFWO and ROUTE_AGWO columns would generally remain as-is, assuming that only SURO was routed through SUSTAIN. At the modeler’s discretion, BMP infiltrated runoff from SUSTAIN can be stored in an aquifer object and routed to the jurished outlet to preserve the water balance. Assuming the linkage file is generated at the most downstream jurished outlet, that portion would be part of the linkage file timeseries output.

Once the linkage controls have been configured, the LSPC model can be executed. The post-implementation timeseries that results from the routing network for this run can be used to evaluate the impact from the upstream BMPs on instream pollutant concentrations and flows.